

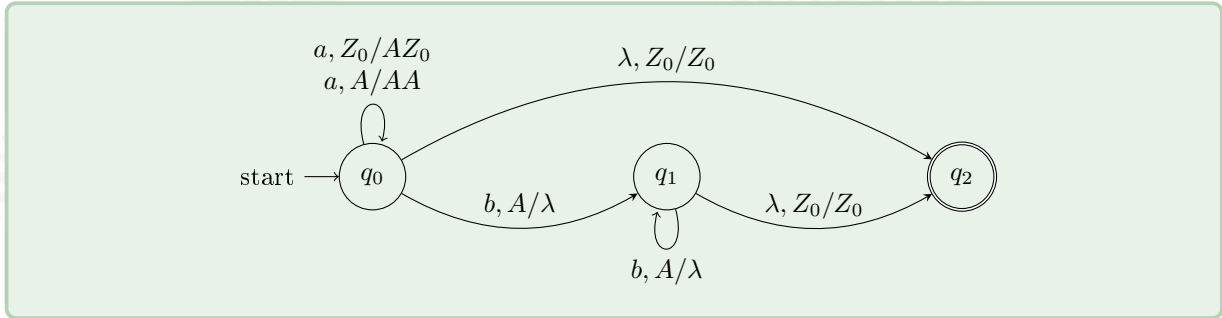
Practica 5: Automatas de Pila

26 de septiembre de 2026

Ejercicio 1. Para cada uno de los siguientes lenguajes, construir un automata de pila que los acepte. Hacer una version deterministica en los casos en que sea posible.

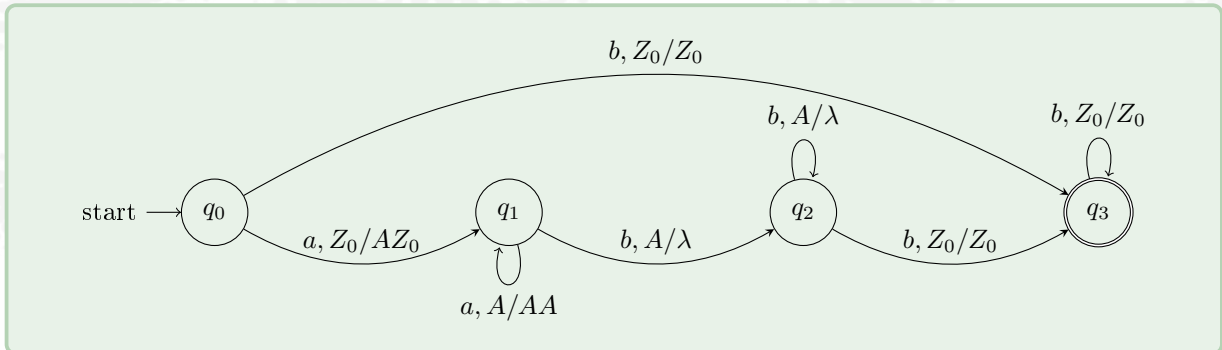
a) $\{a^n b^n\}$

Sol: $\langle \{q_0, q_1, q_2\}, \{a, b\}, \{A, Z_0\}, \delta, q_0, Z_0, \{q_2\} \rangle$



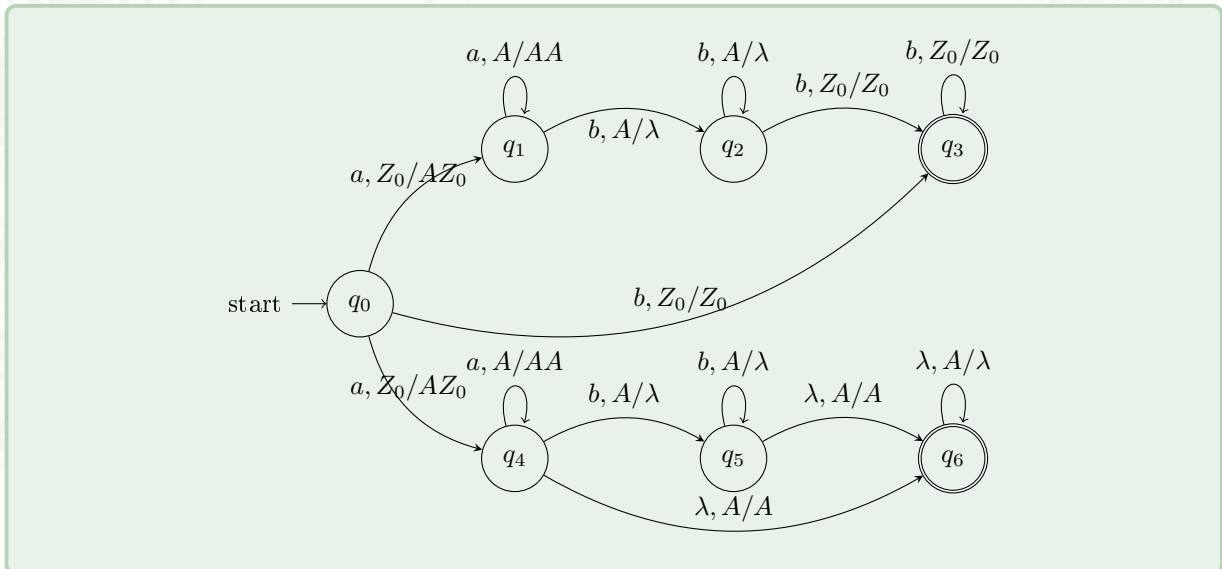
b) $\{a^n b^m \mid m > n\}$

Sol: $\langle \{q_0, q_1, q_2, q_3\}, \{a, b\}, \{A, Z_0\}, \delta, q_0, Z_0, \{q_3\} \rangle$



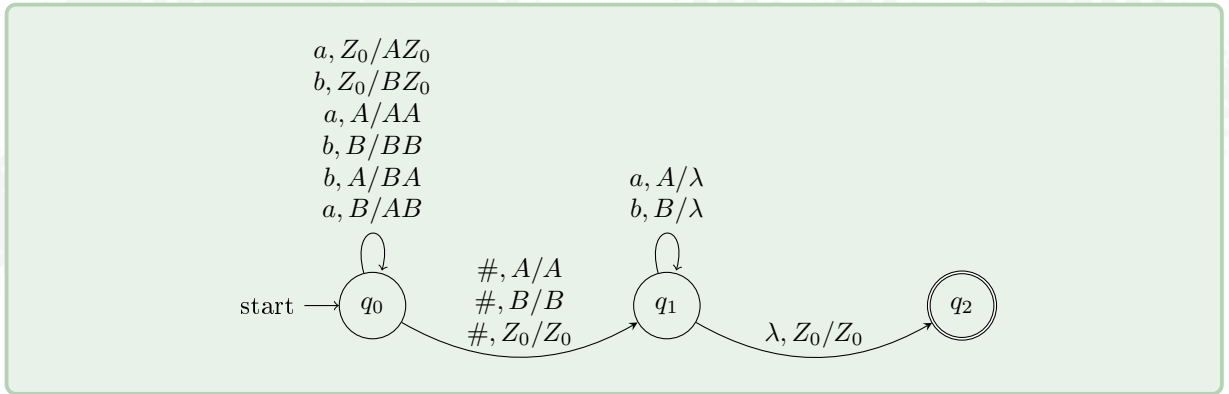
c) $\{a^n b^m \mid m \neq n\}$

Sol: $\langle \{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}, \{a, b\}, \{A, Z_0\}, \delta, q_0, Z_0, \{q_3, q_6\} \rangle$



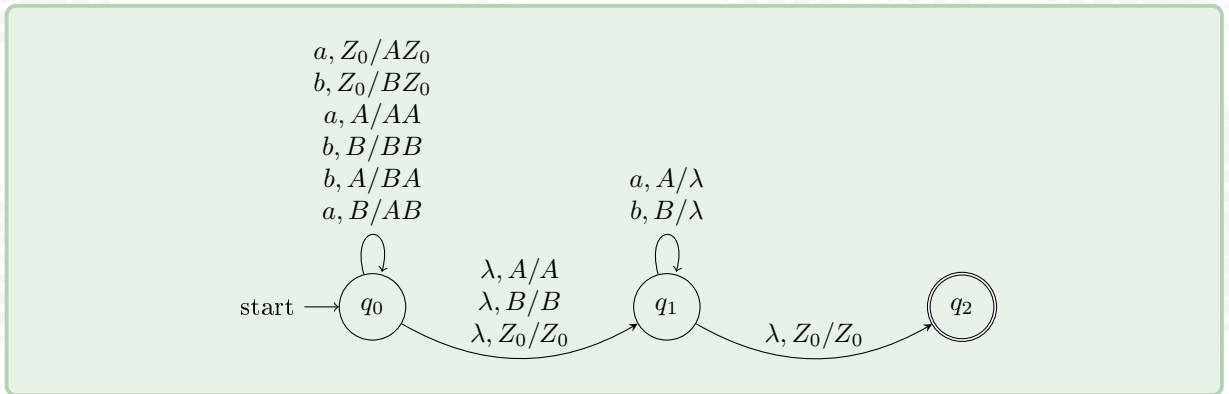
d) $\{\omega \# \omega^r \mid \omega \in \{a, b\}^*\}$

Sol: $\langle \{q_0, q_1, q_2\}, \{a, \#, b\}, \{A, B, Z_0\}, \delta, q_0, Z_0, \{q_2\} \rangle$



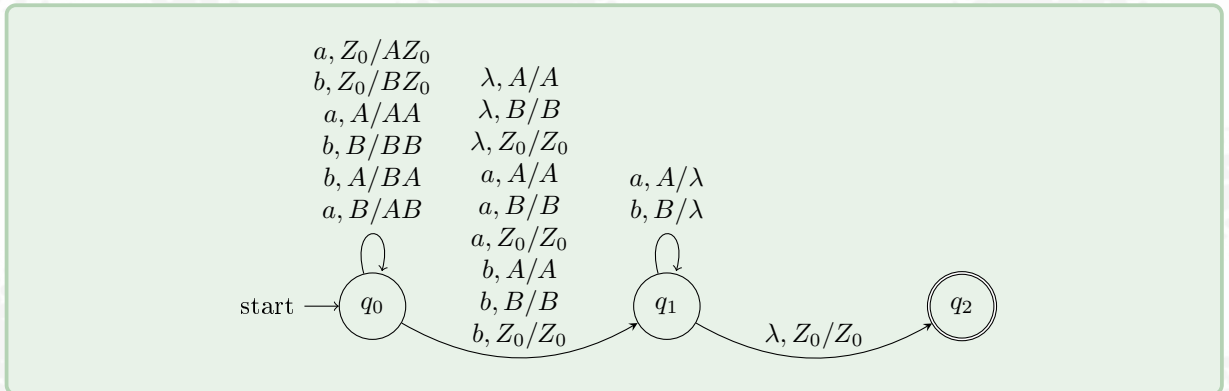
e) $\{\omega \omega^r \mid \omega \in \{a, b\}^*\}$

Sol: $\langle \{q_0, q_1, q_2\}, \{a, b\}, \{A, B, Z_0\}, \delta, q_0, Z_0, \{q_2\} \rangle$



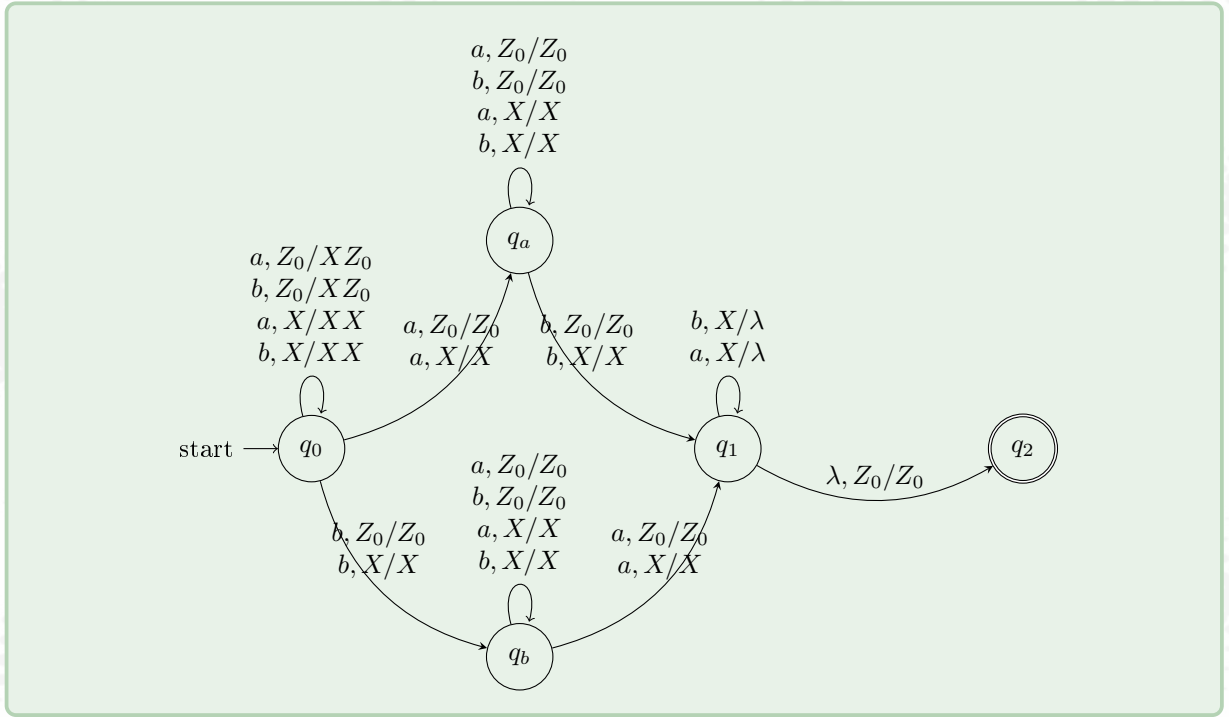
f) $\{\omega \mid \omega \in \{a, b\}^* \wedge \omega = \omega^r\}$

Sol: $\langle \{q_0, q_1, q_2\}, \{a, b\}, \{A, B, Z_0\}, \delta, q_0, Z_0, \{q_2\} \rangle$



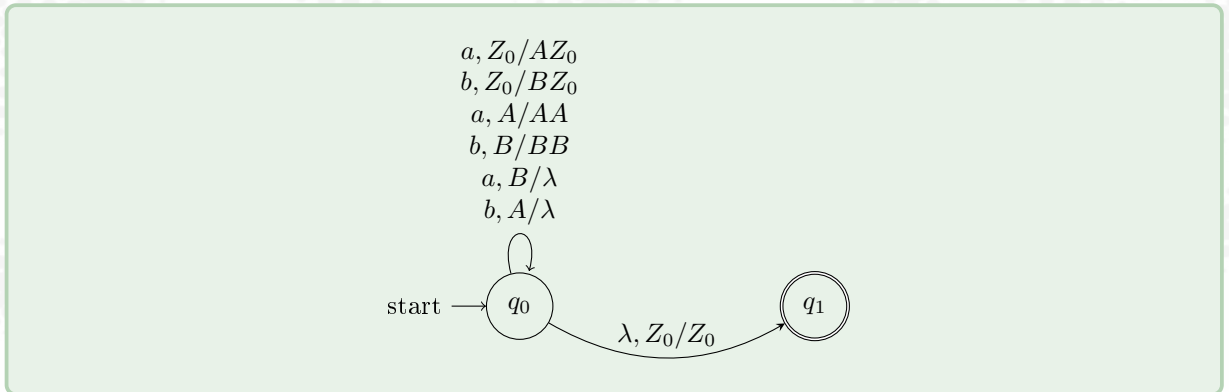
g) $\{\omega \mid \omega \in \{a, b\}^* \wedge \omega \neq \omega^r\}$

Sol: $\langle \{q_0, q_a, q_b, q_1, q_2\}, \{a, b\}, \{X, Z_0\}, \delta, q_0, Z_0, \{q_2\} \rangle$



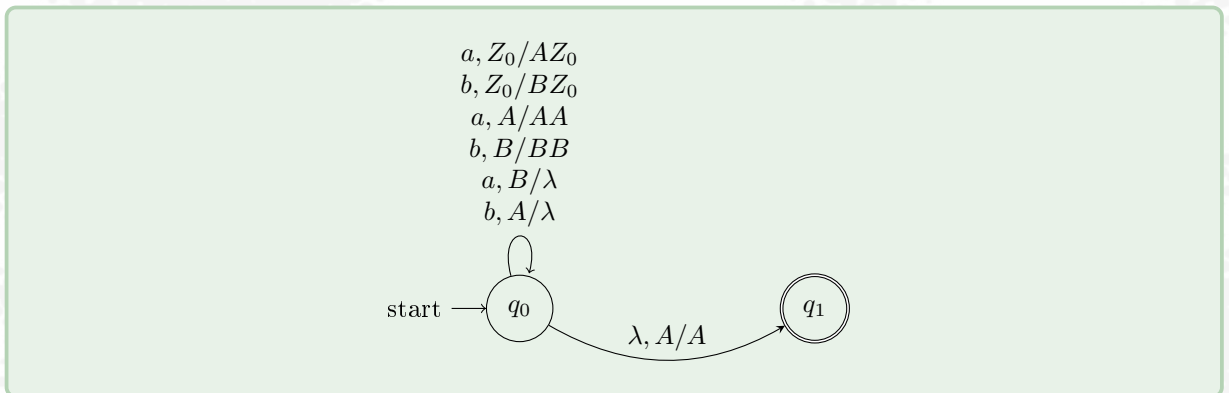
h) $\{\omega \mid \omega \in \{a, b\}^* \wedge \omega_a = \omega_b\}$

Sol: $\langle \{q_0, q_1\}, \{a, b\}, \{A, B, Z_0\}, \delta, q_0, Z_0, \{q_1\} \rangle$



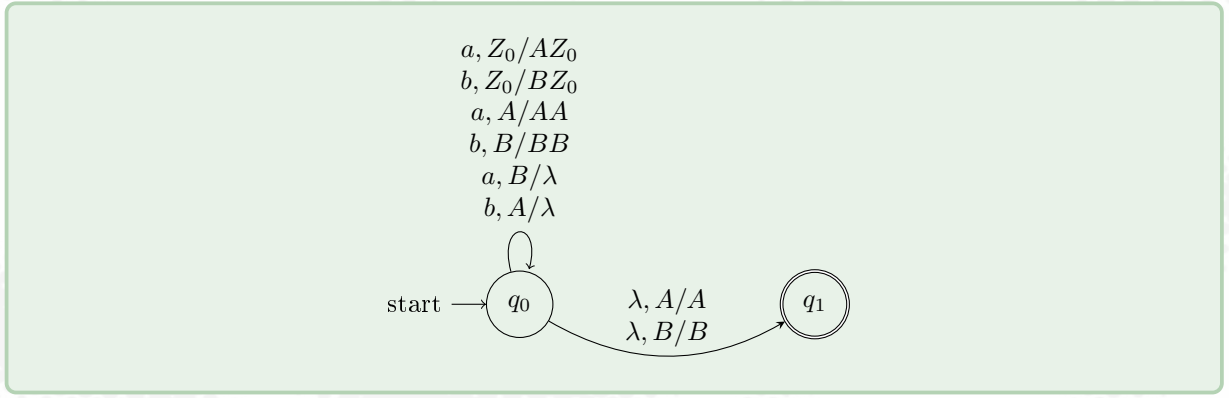
i) $\{\omega \mid \omega \in \{a, b\}^* \wedge \omega_a > \omega_b\}$

Sol: $\langle \{q_0, q_1\}, \{a, b\}, \{A, B, Z_0\}, \delta, q_0, Z_0, \{q_1\} \rangle$



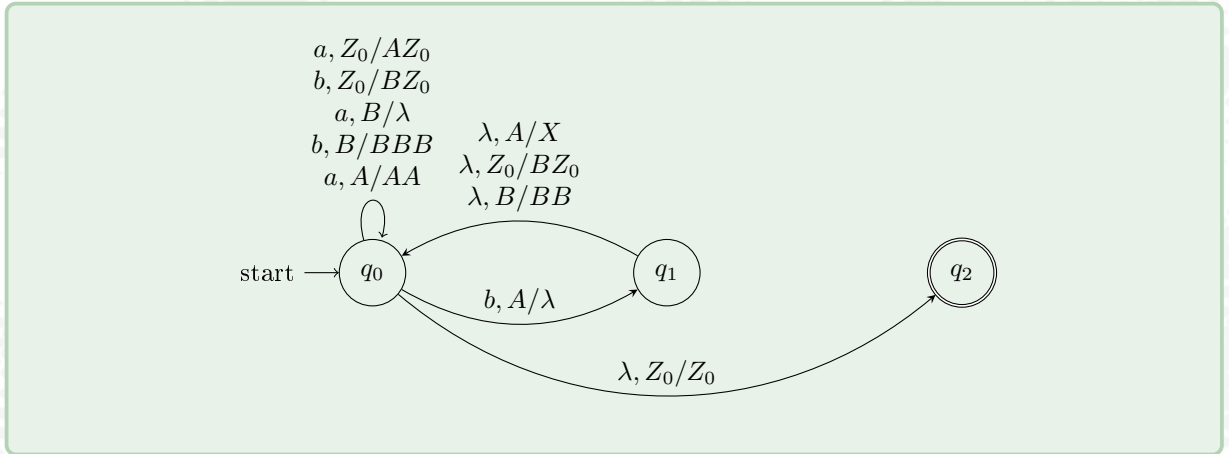
j) $\{\omega \mid \omega \in \{a, b\}^* \wedge \omega_a \neq \omega_b\}$

Sol: $\langle \{q_0, q_1\}, \{a, b\}, \{A, B, Z_0\}, \delta, q_0, Z_0, \{q_1\} \rangle$



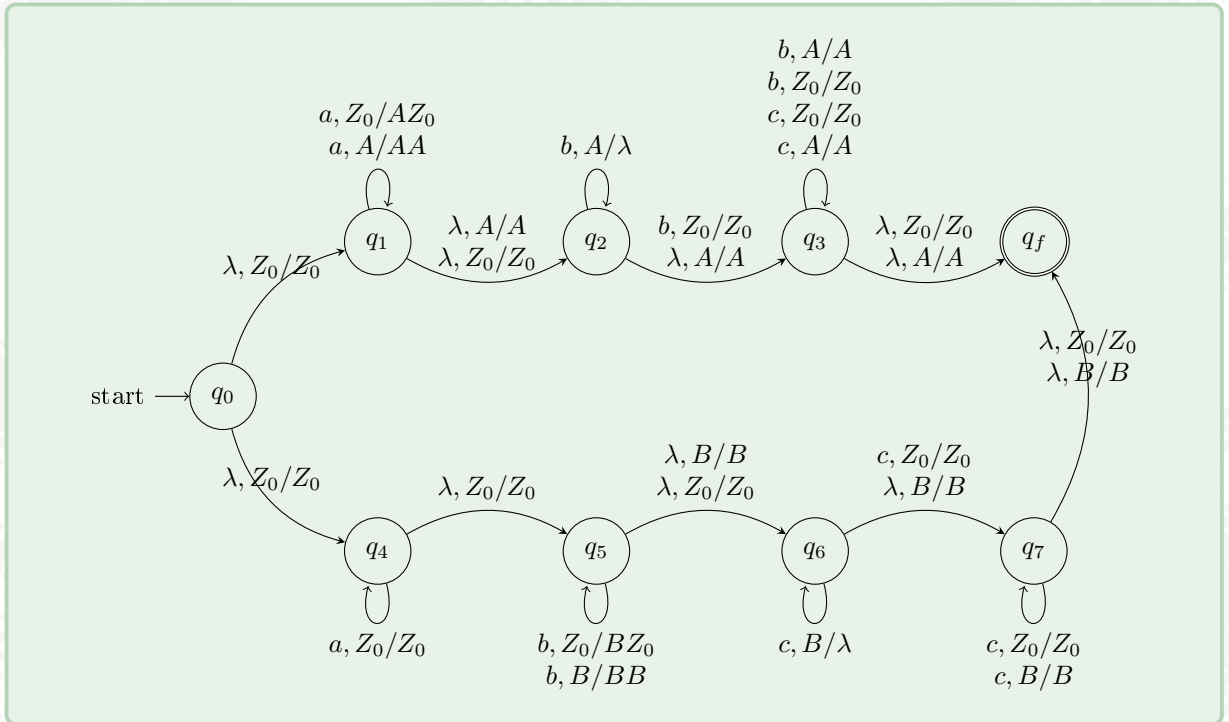
k) $\{\omega \mid \omega \in \{a, b\}^* \wedge \omega_a = 2\omega_b\}$

Sol: $\langle \{q_0, q_1, q_2\}, \{a, b\}, \{A, B, Z_0\}, \delta, q_0, Z_0, \{q_2\} \rangle$



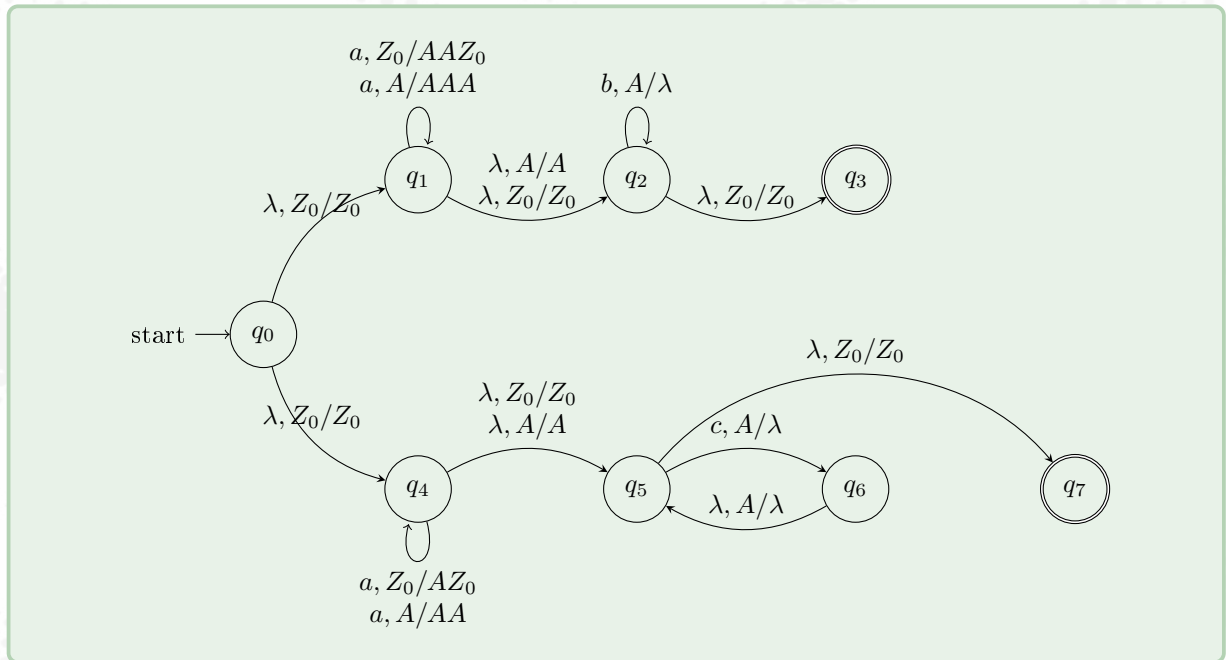
l) $\{a^n b^m c^k \mid n \neq m \vee m \neq k\}$

Sol: $\langle \{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_f\}, \{a, b, c\}, \{A, B, Z_0\}, \delta, q_0, Z_0, \{q_f\} \rangle$



m) $\{a^n b^m \mid m = 2n\} \cup \{a^n c^m \mid n = 2m\}$

Sol: $\langle \{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7\}, \{a, b, c\}, \{A, Z_0\}, \delta, q_0, Z_0, \{q_7\} \rangle$



n) Cadenas de parentesis balanceados.

Sol: $\langle \{q_0, q_1\}, \{(\,,\,)\}, \{X, Z_0\}, \delta, q_0, Z_0, \{q_1\} \rangle$

